Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 09)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 1 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 2 . 1 Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

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0 2 . 2 Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

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0 2 . 3 Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

0 2 . 4 Give **three** other positive feedback mechanisms.

[3 marks]

1

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0 2 . 5 Explain what is meant by a climate **tipping point**.

[2 marks]

15

0 3 Table 1 shows the volume of water in some reservoirs of the hydrological cycle and the annual transfer rates through them.

Table 1

Reservoir	Volume / x 10^{15} dm^3	Annual transfer (inflow = outflow) / x 10^{15} dm^3 yr^{-1}	Percentage of global water / %
Oceans	1 340 000	425	97
Ice caps and glaciers	24 000	3	1.7
Groundwater	23 000	12	1.7
Lakes and rivers	180	40	0.013
Soil moisture	17	80	0.001
Atmosphere	13	480	0.001

0 3 . 1 The residence time of water in a reservoir is given by: residence time = volume in reservoir / annual transfer rate. Use Table 1 to calculate the residence time of water in the atmosphere. Give your answer to the nearest whole day.

[3 marks]

Answer _____ days

0 3 . 2 Calculate the residence time of water in the oceans.

[1 mark]

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Answer [ ] years

0 3 . 3 Explain why the atmosphere is an important reservoir even though it holds only 0.001% of global water.

[2 marks]

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0 3 . 4 Explain why human changes to a small reservoir such as soil moisture are more obvious than changes to the oceans.

[2 marks]

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0 3 . 5 Explain why the hydrological cycle is described as being in dynamic equilibrium.

[2 marks]

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10

0 4 . 1 Describe **three** factors that have restricted the growth of nuclear power.

[3 marks]

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0 4 . 2 Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.

[3 marks]

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0 4 . 3 Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.

[4 marks]

Advantages .....

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Disadvantages .....

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0 5 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 5 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 5 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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0 6 . 1 Explain how each of the following environmental factors affects the rate of degradation of pollutants. Give a named example of a pollutant for each factor.

[6 marks]

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Temperature .....

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Light .....

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Oxygen .....

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0 6 . 2 Explain how the presence of adsorbent materials such as clay can reduce the severity of pollution by heavy metals or radioactive caesium-137.

[2 marks]

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0 6 . 3 Explain why a fall in pH increases the harm caused by many heavy metals.

[2 marks]

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0 7 . 1 Describe **four** factors that should be considered when choosing a site for a dam and reservoir.

[4 marks]

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0 7 . 2 Give **one** advantage and **two** disadvantages of an estuary barrage compared with a reservoir on land.

[3 marks]

Advantage .....

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Disadvantage 1 .....

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Disadvantage 2 .....

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0 7 . 3 Explain why desalination of seawater is only used in some countries.

[2 marks]

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0 7 . 4 Explain how artificial recharge can make aquifer supplies sustainable.

[1 mark]

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0 8 Table 1 shows typical noise levels of some sounds.

Table 1

Sound	Noise level / dB
Threshold of human hearing (at 1000 Hz)	0
Leaves rustling	20
Library	40
Normal conversation	60
Busy road	80
Chain saw	110
Jet engine at 30 m	140

0 8 . 1 Explain why a logarithmic scale is used to measure sound.

[2 marks]

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0 8 . 2 Use Table 1 to calculate how many times greater the sound intensity of a chain saw is than that of a busy road.

[2 marks]

Answer ..... times

0 8 . 3 A machine produces 85 dB. A second identical machine is switched on next to it. State the new noise level and explain your answer.

[2 marks]

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0 8 . 4 Explain what is meant by the statement that 0 dB does not mean there is no sound.

[1 mark]

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0 8 . 5 State the frequency range of hearing for a typical young person and explain why the dB(A) scale is used for human hearing.

[3 marks]

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0 9 Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

Table 1

Feature	Deposit X	Deposit Y
Ore grade / % copper	0.9	0.35
Depth of deposit / m	350	40
Overburden	Hard igneous rock	Loose glacial sediment
Distance to nearest railway / km	120	8
Annual rainfall / mm	2 100	500
Chemical form	Copper sulfide	Copper sulfide

0 9 . 1 Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

[2 marks]

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0 9 . 2 Use Table 1 to discuss the factors that affect the viability of mining each deposit.

[6 marks]

0 9 . 3 State **two** changes that would decrease the cut-off ore grade.

[2 marks]

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2 .....

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1 0 . 1 Explain how the properties of smoke and of the gases involved in acid rain lead to their environmental effects, and evaluate the effectiveness of the methods used to control them.

[9 marks]

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Extra space

1 0 . 2 State **one** effect of tropospheric ozone on plants.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

OR

1 1 . 2 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS